

Operating Systems

What is an Operating System?

An **Operating System (OS)** is system software that manages computer hardware and software and acts as the main interface between the user and the machine. It allows you to run apps, manage files, use memory, connect devices, and keep the whole system organized.

In simple words, the OS is the **boss** of the computer. Without it, your computer or phone cannot function in a useful way.

Why operating systems are important

An OS makes the device usable by handling the background work so users can focus on applications. It manages resources like CPU time, memory, files, storage, and security.

It also provides a platform for programs to run efficiently and safely.

Main functions of an OS

1. Process management

The OS creates, schedules, and stops processes, which are programs in execution. It decides which program gets CPU time and for how long.

2. Memory management

The OS allocates memory to programs and takes it back when they finish. This prevents programs from interfering with one another.

3. File management

The OS organizes files and folders, controls reading and writing, and keeps data structured.

4. Device management

The OS communicates with input/output devices such as keyboards, printers, disk drives, and displays through drivers.

5. Security and access control

The OS protects data and system resources by controlling user access and permissions.

6. User interface

The OS provides a command-line interface or graphical interface so users can interact with the machine easily.

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Types of operating systems

There are several major OS types, each designed for different environments.

Batch operating system

Jobs are collected and processed in groups with little or no user interaction.

Time-sharing operating system

Multiple users or tasks share CPU time so the system feels interactive.

Multiprogramming operating system

Several programs are kept in memory so the CPU stays busy.

Multiprocessing operating system

Uses more than one processor or CPU core to execute tasks in parallel.

Distributed operating system

Multiple computers work together and appear as one system.

Real-time operating system

Designed for fast and predictable response, often used in control systems.

Network operating system

Manages network resources and supports communication between connected machines.

Embedded operating system

Used in devices like routers, appliances, cars, and industrial systems.

Mobile operating system

Designed for smartphones and tablets, such as Android and iOS.

Popular operating systems

Some of the most common operating systems are:

- Windows.
- macOS.
- Linux.
- Android.
- iOS.

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OS architecture basics

An operating system is often described in layers. Hardware sits at the bottom, the kernel manages core operations, and user-facing tools and applications sit above it.

Kernel

The kernel is the core part of the OS. It manages CPU, memory, and device access.

System calls

System calls allow programs to request services from the OS safely.

User space

Applications and utilities run above the kernel and interact with the system through interfaces and libraries.

Linux as an example

Linux is a popular operating system family widely used in servers, cloud systems, development, embedded devices, and many desktops. It is known for being powerful, flexible, and widely used in technical fields.

A Linux system typically includes hardware, the Linux operating system layer, standard libraries, utility programs, and user interfaces.

Important OS concepts

Multitasking

The OS runs multiple applications at the same time by switching CPU attention quickly.

Multithreading

A process can have multiple threads that share resources but perform different tasks.

Scheduling

The OS decides the order and time slices for processes.

Virtual memory

The OS uses disk space to extend memory when physical RAM is limited.

Deadlock

Deadlock happens when processes wait forever for resources held by each other.

File permissions

Permissions control who can read, write, or execute a file.

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Real-world uses

Operating systems are everywhere:

- Computers.
- Smartphones.
- Smart TVs.
- Cars.
- Smart appliances.
- ATMs.
- Industrial machines.
- Servers and cloud systems.

How to study OS from zero

Step 1: Learn the basics

Understand what an OS is, why it exists, and what problems it solves.

Step 2: Learn core functions

Study process management, memory management, file systems, and device management.

Step 3: Learn process concepts

Understand process states, scheduling, context switching, and threads.

Step 4: Learn memory management

Study paging, segmentation, virtual memory, and allocation methods.

Step 5: Learn file systems and security

Understand directories, permissions, access control, and storage organization.

Step 6: Learn advanced topics

Study deadlocks, synchronization, disk scheduling, and virtualization.

Exam-ready short notes

Definition

An OS is system software that manages hardware and software resources and provides services to programs and users.

Main functions

Process management, memory management, file management, device management, and security.

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Examples

Windows, Linux, macOS, Android, and iOS.

Goal of OS

To make the computer efficient, secure, and easy to use.

Small project ideas related to OS

If you want to build projects while learning OS, try:

- CPU scheduling simulator.
- Memory allocation simulator.
- File system explorer.
- Process manager dashboard.
- Simple shell in C.
- Deadlock detection demo.
- Page replacement algorithm visualizer.

Final summary

Operating systems are the foundation of all modern computing. They manage hardware, run applications, protect data, and provide the environment that makes computers and phones useful. If you understand OS well, you build a strong base for programming, Linux, DevOps, system design, and cybersecurity.